Lecture 13 (Even More Simplex Method)

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We will work through an example of the simplex method and briefly talk about O notation today.

Simplex Method

Pivoting Rules

We have been ignoring degenerate BFS so far. What might happen if we have a degenerate BFS? We have a few possibilities

a) If the current BFS, $\vec{x}$, is degenerate, then θ^* could be 0. In this case, out new BFS $\vec{y} = \vec{x} + \theta^* \vec{d} = \vec{x}$. We can still update B though. This will replace one of the columns corresponding to a variable with value, 0, with the column A_j.

b) If we start with a non-degenerate solution, with $\theta^* > 0$, it may be the case that $\vec{x} + \theta^* \vec{d}$ will have multiple basic variables go to 0. Only one can exit the basis.

If we are not careful, we could end up getting into infinite loops with these degenerate solutions. These infinite loop behavior is called, **cycling**. In practice, this doesn't happen very often, but when it does, you can waste a lot of computation time and resources trying to resolve this. To avoid cycling, we can use pivoting rules. The rules for picking which variable with a negative reduced cost to activate and for picking which variable with the minimum θ^* are called **pivoting rules**. Here are some common ones:

Rule 1

Activate x_j is $\bar{c}_j$ is the most negative of all negative reduced costs.

Strengths

- Relatively easy to compute
- Moves in the direction of steepest descent

Weaknesses

- We don't know how far we can move in this direction

Rule 2

Activate x_j with $\bar{c}_j < 0$ and such that $\theta^* |\bar{c}_j|$ is the largest.

Strengths

- Gives the largest possible decrease in the objective function in each step.

Weaknesses

- Costly to compute
- Comparable in runtime to **Rule 1** in practice.

Rule 3

Activate x_j with the smallest j of all the j such that $\bar{c}_j < 0$.

Strengths

- Quickest to compute (you only need to compute reduce costs until you find the first negative reduced cost)
- If you use this lowest index rule for selecting variables to activate/deactivate, you avoid cycling (We may prove this in a future class)

There are other more complicated rules that some solvers use. These more complicated rules may converge to the solution faster, but are also harder to implement. We won't discuss these in this class.

An Example of the Simplex Method

Let's go through a full example of the simplex method for practice. Consider the LP below:

$$\text{minimize } 2x_1 + 4x_2 - 4x_3 + 3x_4 + x_5$$

subject to

$$x_1 + x_2 + x_3 + x_4 + x_5 = 10$$

$$x_1 + 2x_2 + 3x_3 + 0x_4 + x_5 = 16$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

This LP yields the following matrices and vectors:

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 0 & 1 \end{pmatrix},\ \vec{b} = \begin{pmatrix} 10 \\ 16 \end{pmatrix},\ \vec{c} = \begin{pmatrix} 2 \\ 4 \\ -4 \\ 3 \\ 1 \end{pmatrix}$$

Since there are two constraints, we will only activate 2 variables at a time. Suppose we start by trying to select x_1, x_5. Then

$$B = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Notice that B is not invertible. This means we will not get a BFS with x_1, x_5 selected. In fact, there is no feasible solution with x_1 and x_5 active at the same time for this problem. We often need to be careful when picking which constraints to be active. Suppose we pick x_2, x_5 instead. Then

$$B = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix},\ B^{-1} = \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix}$$

So

$$\vec{x}_B = B^{-1}\vec{b} = \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 10 \\ 16 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$

So we have a BFS

$$\vec{x} = \begin{pmatrix} 0 \\ 6 \\ 0 \\ 0 \\ 4 \end{pmatrix}$$

with

$$\vec{c}_B = \begin{pmatrix} 4 \\ 1 \end{pmatrix},\ B^{-1} = \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix}$$

Now let's compute the reduced costs:

$$\bar{c}_1 = c_1 - \vec{c}_B \cdot B^{-1}A_1 = 2 - \begin{pmatrix} 4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 2 - \begin{pmatrix} 4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1$$

$$\bar{c}_2 = 0$$

as x_2 is active.

$$\bar{c}_3 = c_3 - \vec{c}_B \cdot B^{-1}A_3 = -4 - \begin{pmatrix} 4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = -4 - \begin{pmatrix} 4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \end{pmatrix} = -11$$

$$\bar{c}_4 = c_4 - \vec{c}_B \cdot B^{-1}A_4 = 3 - \begin{pmatrix} 4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 3 - \begin{pmatrix} 4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 2 \end{pmatrix} = 5$$

$$\bar{c}_5 = 0$$

as x_5 is active. Only $j = 3$ has a negative reduced cost, so we will select this variable to active. Now, we can compute the $\vec{d}$. For the inactive indices, we have $d_i = 0$. Since we are activating $j = 3$, we see $d_3 = 1$. For the currently active indices, we need to find

$$\vec{d}_B = -B^{-1}A_3$$

We already computed $B^{-1}A_3$ when finding the reduced cost however.

$$\vec{d}_B = -\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

Now, we compute θ^*

$$\min_{\{i=1,\ldots,m | d_{B(i)} < 0\}} \left(-\frac{x_{B(i)}}{d_{B(i)}} \right) = -\frac{6}{-2} = 3$$

Thus

$$\vec{y} = \vec{x} + \theta^* \vec{d} = \begin{pmatrix} 0 \\ 6 \\ 0 \\ 0 \\ 4 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ -2 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 3 \\ 0 \\ 7 \end{pmatrix}$$

Notice that x_2 deactivated and x_3 activated. Now, we can update our B matrix.

$$B = \begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix}$$

Now, we will go through the next pivot.

$$B^{-1} = \begin{pmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$$

So

$$\vec{x}_B = \begin{pmatrix} 3 \\ 7 \end{pmatrix}, \ \vec{c}_B = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$$

Now, let's find the reduced costs for the non-basic variables.

$$\bar{c}_1 = c_1 - \vec{c}_B \cdot B^{-1}A_1 = 2 - \begin{pmatrix} -4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 2 - \begin{pmatrix} -4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1$$

$$\bar{c}_2 = c_2 - \vec{c}_B \cdot B^{-1}A_2 = 4 - \begin{pmatrix} -4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 4 - \begin{pmatrix} -4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \frac{11}{2}$$

$$\bar{c}_4 = c_4 - \vec{c}_B \cdot B^{-1}A_4 = 3 - \begin{pmatrix} -4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 3 - \begin{pmatrix} -4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -\frac{1}{2} \\ \frac{3}{2} \end{pmatrix} = -\frac{1}{2}$$

So we select $j = 4$ to activate.

$$\vec{d}_B = -B^{-1}A_4 = -\begin{pmatrix} -\frac{1}{2} \\ \frac{3}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -\frac{3}{2} \end{pmatrix}$$

Thus

$$\theta^* = -\frac{7}{-\frac{3}{2}} = \frac{14}{3}$$

Thus

$$\vec{y} = \vec{x} + \theta^*\vec{d} = \begin{pmatrix} 0 \\ 0 \\ 3 \\ 0 \\ 7 \end{pmatrix} + \frac{14}{3}\begin{pmatrix} 0 \\ 0 \\ \frac{1}{2} \\ 1 \\ -\frac{3}{2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \frac{16}{3} \\ \frac{14}{3} \\ 0 \end{pmatrix}$$

Now, we update our B matrix

$$B = \begin{pmatrix} 1 & 1 \\ 3 & 0 \end{pmatrix}$$

So

$$\vec{x}_B = \begin{pmatrix} \frac{16}{3} \\ \frac{14}{3} \end{pmatrix},\ \vec{c}_B = \begin{pmatrix} -4 \\ 3 \end{pmatrix},\ B^{-1} = \begin{pmatrix} 0 & \frac{1}{3} \\ 1 & -\frac{1}{3} \end{pmatrix}$$

Now we find the reduced costs.

$$\bar{c}_1 = c_1 - \vec{c}_B \cdot B^{-1}A_1 = 2 - \begin{pmatrix} -4 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 0 & \frac{1}{3} \\ 1 & -\frac{1}{3} \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 2 - \begin{pmatrix} -4 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \end{pmatrix} = 2 - \frac{2}{3} = \frac{4}{3}$$

$$\bar{c}_2 = c_2 - \vec{c}_B \cdot B^{-1}A_2 = 4 - \begin{pmatrix} -4 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 0 & \frac{1}{3} \\ 1 & -\frac{1}{3} \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix} = 4 - \begin{pmatrix} -4 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} \frac{2}{3} \\ \frac{1}{3} \end{pmatrix} = \frac{17}{3}$$

$$\bar{c}_5 = c_5 - \vec{c}_B \cdot B^{-1}A_5 = 1 - \begin{pmatrix} -4 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 0 & \frac{1}{3} \\ 1 & -\frac{1}{3} \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 - \begin{pmatrix} -4 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \end{pmatrix} = 1 - \frac{2}{3} = \frac{1}{3}$$

As all of the reduced costs are positive, then $\vec{x} = \begin{pmatrix} 0 \\ 0 \\ \frac{16}{3} \\ \frac{14}{3} \\ 0 \end{pmatrix}$ is optimal and the algorithm terminates.

Big O Notation

Let $f, g : \mathbb{R} \to \mathbb{R}$ be two real valued functions. We say $f(x) = O(g(x))$ if there exists $C, N > 0$ such that

$$|f(x)| \leq C\,|g(x)|$$

for all $x \geq N$. Oftentimes, we will require f, g be integer valued instead. This is usually the case when we use f, g to reference the number of operations an algorithm takes (as a function of the sizes of the inputs). Let's give some examples of this notation.

Ex 1

Let $f(x) = 3x$, $g(x) = 2x$. Then notice we can set $C = 3$, then

$$|f(x)| = |3x| \leq |6x| = 3\,|2x| = 3\,|g(x)|$$

for all $x \geq 1 = N$ (note we need to set some N for this to be true).

Ex 2

Let $f(x) = x + 3$, $g(x) = x^2$. Notice at $x = 0$ we have

$$f(0) = 3 > 0 = g(0)$$

however, if we pick a large N (maybe $N = 6$), we see

$$f(6) = 9,\ g(6) = 36$$

so

$$|f(6)| \leq 1\,|g(6)|$$

and as $f'(x) = 1$ but $g'(x) = 2x$ then for all $x \geq 6$, we see

$$f'(x) < g'(x)$$

thus

$$|f(x)| \leq |g(x)|$$

for all $x \geq 6$.

Operations

Typically we only count multiplications and divisions of scalars for the purposes of operations. We will assume that addition/subtraction can be done in constant time. We can talk about the complexity of various matrix/vector operations.

Ex 1

Consider $\vec{x}, \vec{y} \in \mathbb{R}^n$. Then the operation $\vec{x} \cdot \vec{y}$ will require $O(n)$ operations as

$$\vec{x} \cdot \vec{y} = x_1 y_1 + \ldots x_n y_n$$

Each term in the sum yields a separate multiplication operation. We assume the sum operation can be done with negligible time.

Ex 2

Consider $A \in \mathbb{R}^{m \times n}$ and $\vec{b} \in \mathbb{R}^n$. Then the operation $A\vec{b}$ will require $O(mn)$ operations as

$$A\vec{b} = \begin{pmatrix} A_1 \cdot \vec{b} \\ \vdots \\ A_m \cdot \vec{b} \end{pmatrix}$$

where each A_i is the ith row of A.

Ex 3

I claim (without proof) that finding A^{-1} for $A \in \mathbb{R}^{n \times n}$ is $O\left(n^3\right)$. This means in the simplex method we will see a bottleneck for the step where we find B^{-1}. When we talk about the refined simplex method, we will find a way to get B^{-1} without the full $O\left(n^3\right)$ steps.